

Samuel Eder

AI & MUSIC PATTERN RECOGNITION SPECIALIST

BA Instrumental Pedagogy (Mozarteum) · BSc AI in progress (JKU)
MA Clarinet (exp. Aug 2026)



+43 680 1416729
samuel.eder@live.at
linkedin.com/in/samuel-eder
github.com/Walsamer
huggingface.co/clarsam
walsamer.github.io/samuel_eder_cv
Vienna, Austria

CORE SKILLS

Audio Pattern Recognition

ML Model Training

Dataset Design & QA

Project Management

Python PyTorch

TensorFlow Deep Learning

Data Operations

LLM Evaluation

Workflow Automation

Quality Control

Cross-team Coordination

NLP Data Governance

MUSIC EXPERTISE

Clarinet (Master level)

Saxophone Music Notation

Instrument Pedagogy

Audio Signal Processing

Pitch & Rhythm Analysis

Music Theory & Composition

Audio Production & Mixing

Logic Pro

Real-time Audio Processing

LANGUAGES

German **Native**

English **Fluent**

TOOLS & TECH

Python / PyTorch / TensorFlow

Shiny for Python

CNN Architectures

Real-time Audio Processing

Data Pipelines

Annotation Workflows

GitHub

PROFILE

I sit at the exact intersection this role requires: a **Master's-level musician** (Clarinet, MUK Vienna; Instrumental Pedagogy, Mozarteum Salzburg) who also builds **audio-pattern recognition systems running in real-time on edge devices**. As founder of Insame GmbH I designed the full ML pipeline — from raw audio dataset curation through CNN model training to real-time pitch detection in a mobile app. I combine that hands-on technical depth with three years of leading cross-functional teams across data, engineering, and product, and am currently strengthening my formal AI foundations (BSc, JKU Linz) while working in Data & AI at Magenta Telekom.

Why MyGroove: Shaping pattern recognition systems that are musically meaningful — not just technically correct — is exactly the problem I have spent the last four years solving. The combination of deep music domain knowledge and practical ML experience lets me evaluate model quality the way a musician and an engineer would simultaneously.

PROFESSIONAL EXPERIENCE

Founder & CEO / AI Project Lead

2021 – 2024

Insame GmbH

- Designed and built a **pitch recognition ML model** (CNN, TensorFlow) for real-time audio analysis in a mobile rhythm game
- Designed and curated a custom audio dataset; managed the full annotation, training, and validation lifecycle
- Built internal testing tools to validate model behavior across instruments and edge cases
- Identified and prioritised bugs; maintained model consistency across iterative releases
- Coordinated across data, engineering, and product roles — sole responsible for timelines and quality gates
- Documented all workflows and analysis steps for reproducibility
- Owner of all Insame IP: brand, method books, PDFs, and play-along materials

Data Governance (Internship) – Data & AI

2025 – present

Magenta Telekom / T-Mobile Austria GmbH

- Support ongoing data governance initiatives by helping structure and document data assets across departments
- Assist in analysing reporting requirements, identifying data quality issues, and proposing improvements to data handling processes
- Contribute to internal projects focused on improving transparency and accessibility of business-relevant datasets
- Collaborate with cross-functional teams (analytics, IT, business units, legal) to align data standards and support compliance with internal data policies
- Participate in workshops and internal communication efforts to raise awareness of data-driven practices within the organisation
- Gained practical experience working with enterprise data structures, metadata, and reporting frameworks in a large-scale corporate environment

Marketing & Communication Specialist

2023 – 2024

Maxton GmbH

- Developed and deployed automated workflow pipelines, reducing manual overhead
- Managed cross-channel communication strategies and stakeholder alignment

Clarinet & Saxophone Teacher

2017 – 2025

Musikschule Gänserndorf · Musikschule Walgau · Hiltensweiler · Weißensberg

- Over 8 years of individual and group instruction across Austria and Germany
- Developed tailored lesson plans for all skill levels and ages; prepared students for recitals, examinations, and ensemble performance
- Deep practical understanding of musical technique, methodical teaching and learning of wind instruments, notation, composition, music theory and performance feedback

SELECTED PROJECTS

Pitch Recognition ML Model

Fine-tuned CNN for real-time pitch detection; custom audio dataset design, training splits, TensorFlow deployment in a live mobile app. Validated accuracy across instruments and edge cases.

LLM as a Judge – Research Paper ↗

Self-Preference Bias in LLM Juries: A Controlled Test of Hard Holdout — JKU Linz, 2026. Curated 4,196 judge votes across a 7-model jury; tested hard holdout as a bias mitigation. Finding: vote-level self-preference exists in 6/7 models yet produces no detectable jury-level effect.

Datasurgeon – Data Cleaning Tool

Python/Shiny web app for preprocessing and exploratory analysis of unstructured data. Integrated LLM component for natural-language interaction with tabular datasets.

ZahnML – Clinical NLP Pipeline

End-to-end pipeline for transcription, classification, and structured information extraction from dental inputs. Automated generation of Heil- und Kostenpläne and Materialabrechnungen directly from transcription.

ORCHESTRA & CHAMBER MUSIC

Regular substitute Clarinet / E-flat / Bass Clarinet at **Symphonieorchester Vorarlberg** (2019–2022) and **Wiener Symphoniker** incl. Bregenzer Festspiele (2019–2022); substitute principal clarinet, Citylight Symphonieorchester Luzern (2019). Chamber music: **Chroma Quintet** — world premiere of Pierre Thilloy's *Inferno* (dedicated to the ensemble), Jastrzębie-Zdrój 2025, Lublin 2024, Caylus Festival France 2026 — youtu.be/bb964r8xj1E; Ensemble Ars Braemia (SRG SSR recording 2019), PulsArt Ensemble (ORF Broadcast 2018–2019), Klezmertrio «dri kindsköpf», Trio Astarte. Jazz album *Chroma* with Full Crimp Ensemble.

EDUCATION

Bachelor of Science — Artificial Intelligence (in progress, 4th semester)

Johannes Kepler Universität Linz
2024 – present

Master of Arts — Clarinet (thesis pending, expected Aug 2026)

Musik und Kunst Universität der Stadt Wien (MUK)
2020 – 2026

Bachelor of Arts — Instrumental & Vocal Pedagogy

Universität Mozarteum Salzburg
2015 – 2019

Entrepreneurial Leadership Program

Austrian Startups
2024 – 2025 · National selection; startup finance, product-market fit, team leadership